

GridWise — সম্পূর্ণ Test Case Pack

Service যাচাই করার সব case একসাথে: official ১০টা public sample case, প্রতিটা directive-এর জন্য একটা নতুন ভাষার case, একটা multi-note case, আর error-handling case গুলো। প্রতিটার জন্য হুবহু paste করার body আর হুবহু প্রত্যাশিত ফল দেওয়া আছে।

Live API	https://brian-sheffield-surely-fly.trycloudflare.com
Repository	https://github.com/meherabmehu/GridWise-LLM-Energy-Optimizer
Cases	19 request cases + 3 error cases

১. যেকোনো case চালানোর নিয়ম

1. Chrome-এ <https://brian-sheffield-surely-fly.trycloudflare.com/docs> খোলো।
2. POST /optimize-energy খুলে **Try it out** চাপো।
3. Request body বাক্স খালি করে case-এর body paste করো, তারপর **Execute** চাপো।
4. Response code 200 কি না দেখো, তারপর নিচের সংখ্যাগুলোর সাথে মিলাও।

19টা paste-ready body আছে এই PDF-এর পৃষ্ঠা 6 থেকে 24 পর্যন্ত।

২. Directive coverage

Problem statement-এ বলা ছয়টা directive-ই দুইবার cover করা আছে — একবার official sample case দিয়ে, আরেকবার নতুন ভাষার case দিয়ে।

Directive	official case	নতুন ভাষার case
solar_reduction	SAMPLE-01, SAMPLE-06, SAMPLE-09	P1
minimum_battery_reserve	SAMPLE-03, SAMPLE-07, SAMPLE-10	P2
no_charge_window	SAMPLE-02, SAMPLE-06, SAMPLE-08	P3
no_discharge_window	SAMPLE-04, SAMPLE-08	P4
max_grid_window	SAMPLE-05, SAMPLE-07, SAMPLE-10	P5
no_op	SAMPLE-01, SAMPLE-06, SAMPLE-09, SAMPLE-10	P6

৩. সব case

1. SAMPLE-01 official SAMPLE-01 - official public sample case

- Facilities will wash the rooftop solar panels from noon until 2 PM. During cleaning, usable solar should be treated as roughly 25% of the forecast.
- The sports office moved next month's registration deadline.

প্রত্যাশিত: solar_reduction (hours [12, 13], factor 0.25) | no_op

মোট: 2692.50 kWh / 38365.00 BDT / peak 175.00 kWh

paste করার body: এই PDF-এর পৃষ্ঠা 6

2. SAMPLE-02 official SAMPLE-02 - official public sample case

- The battery charger will be isolated from 2 AM until 5 AM for electrical maintenance.

প্রত্যাশিত: no_charge_window (hours [2, 3, 4])

মোট: 2915.00 kWh / 42885.00 BDT / peak 180.00 kWh

paste করার body: এই PDF-এর পৃষ্ঠা 7

3. SAMPLE-03 official SAMPLE-03 - official public sample case

- Keep at least 50% of the battery capacity stored in the battery from 6 PM until 9 PM for emergency operations.

প্রত্যাশিত: minimum_battery_reserve (hours [18, 19, 20], minimum_energy_kwh 100)

মোট: 2430.00 kWh / 35480.00 BDT / peak 205.00 kWh

paste করার body: এই PDF-এর পৃষ্ঠা 8

4. SAMPLE-04 official SAMPLE-04 - official public sample case

- For protection testing, the battery must not discharge from 6 PM until 8 PM.

প্রত্যাশিত: no_discharge_window (hours [18, 19])

মোট: 2645.00 kWh / 40495.00 BDT / peak 225.00 kWh

paste করার body: এই PDF-এর পৃষ্ঠা 9

5. SAMPLE-05 official SAMPLE-05 - official public sample case

- From 6 PM until 9 PM, campus grid import must not exceed 155 kWh in any hour because the feeder is operating under a temporary limit.

প্রত্যাশিত: max_grid_window (hours [18, 19, 20], max_grid_kwh 155)

মোট: 2430.00 kWh / 33950.00 BDT / peak 175.00 kWh

paste করার body: এই PDF-এর পৃষ্ঠা 10

6. SAMPLE-06 official SAMPLE-06 - official public sample case

- Cloud cover during panel inspection will leave about half of the forecast solar output from 10 AM until noon.
- The charging circuit will be unavailable from 2 PM until 4 PM.
- The library is extending book-return hours next week.

প্রত্যাশিত: solar_reduction (hours [10, 11], factor 0.5) | no_charge_window (hours [14, 15]) | no_op

মোট: 2395.00 kWh / 34090.00 BDT / peak 175.00 kWh

paste করার body: এই PDF-এর পৃষ্ঠা 11

7. SAMPLE-07 **official** SAMPLE-07 - official public sample case

- Keep at least 90 kWh in the battery from 6 PM until 10 PM for emergency services.
- The evening transformer limit is 180 kWh of grid import from 7 PM until 9 PM.

প্রত্যাশিত: minimum_battery_reserve (hours [18, 19, 20, 21], minimum_energy_kwh 90) | max_grid_window (hours [19, 20], max_grid_kwh 180)

মোট: 2560.00 kWh / 38550.00 BDT / peak 185.00 kWh

paste করার body: এই PDF-এর পৃষ্ঠা 12

8. SAMPLE-08 **official** SAMPLE-08 - official public sample case

- Battery charging is disabled from 11 AM until 1 PM while technicians inspect the charger.
- Do not discharge the battery from 5 PM until 7 PM during relay testing.

প্রত্যাশিত: no_charge_window (hours [11, 12]) | no_discharge_window (hours [17, 18])

মোট: 2490.00 kWh / 37665.00 BDT / peak 210.00 kWh

paste করার body: এই PDF-এর পৃষ্ঠা 13

9. SAMPLE-09 **official** SAMPLE-09 - official public sample case

- Expect an 80% reduction in rooftop solar between 11 AM and 2 PM because of inverter work.
- The student affairs office will publish club notices tomorrow.

প্রত্যাশিত: solar_reduction (hours [11, 12, 13], factor 0.2) | no_op

মোট: 2504.00 kWh / 34873.00 BDT / peak 170.00 kWh

paste করার body: এই PDF-এর পৃষ্ঠা 14

10. SAMPLE-10 **official** SAMPLE-10 - official public sample case

- The data center requires at least 80 kWh to remain in the battery from 6 PM until 10 PM.
- Grid intake must stay at or below 190 kWh from 7 PM until 10 PM while the substation is constrained.
- A seminar room booking was moved to next week.

প্রত্যাশিত: minimum_battery_reserve (hours [18, 19, 20, 21], minimum_energy_kwh 80) | max_grid_window (hours [19, 20, 21], max_grid_kwh 190) | no_op

মোট: 2715.00 kWh / 41620.00 BDT / peak 190.00 kWh

paste করার body: এই PDF-এর পৃষ্ঠা 15

11. P1 **paraphrase** solar_reduction - ৫৫% হ্রাস, সকাল ৯টা থেকে ১১টা

- Expect a 55% reduction in rooftop solar from 9 AM to 11 AM.

প্রত্যাশিত: solar_reduction, hours [9, 10], factor 0.45

মোট: 2551.0 kWh / 35757.00 BDT / peak 175.0 kWh

paste করার body: এই PDF-এর পৃষ্ঠা 16

12. P2 **paraphrase** minimum_battery_reserve - অন্তত ১১৫ kWh, সন্ধ্যা ৭টা থেকে ১০টা

- Keep at least 115 kWh in the battery from 7 PM until 10 PM.

প্রত্যাশিত: minimum_battery_reserve, hours [19, 20, 21], minimum_energy_kwh 115.0

মোট: 2430.0 kWh / 35190.00 BDT / peak 200.0 kWh

paste করার body: এই PDF-এর পৃষ্ঠা 17

13. P3 **paraphrase** no_charge_window - চার্জ বন্ধ, রাত ২টা থেকে ৫টা

- Please don't charge the battery between 2 AM and 5 AM.

প্রত্যাশিত: no_charge_window, hours [2, 3, 4]

মোট: 2430.0 kWh / 34130.00 BDT / peak 175.0 kWh

paste করার body: এই PDF-এর পৃষ্ঠা 18

14. P4 **paraphrase** no_discharge_window - ডিসচার্জ বন্ধ, বিকাল ৪টা থেকে ৬টা

- Battery discharge is prohibited between 4 PM and 6 PM during relay work.

প্রত্যাশিত: no_discharge_window, hours [16, 17]

মোট: 2430.0 kWh / 34070.00 BDT / peak 175.0 kWh

paste করার body: এই PDF-এর পৃষ্ঠা 19

15. P5 **paraphrase** max_grid_window - import সর্বোচ্চ ১৫৫ kWh, সন্ধ্যা ৬টা থেকে ৯টা

- Between 6 PM and 9 PM the campus feeder is limited, so keep hourly grid import under 155 kWh.

প্রত্যাশিত: max_grid_window, hours [18, 19, 20], max_grid_kwh 155.0

মোট: 2430.0 kWh / 33950.00 BDT / peak 175.0 kWh

paste করার body: এই PDF-এর পৃষ্ঠা 20

16. P6 **paraphrase** no_op - যে note বাদ দিতে হবে

- The library extended its opening hours during exam week.

প্রত্যাশিত: no_op, applies false, structured_adjustment null

মোট: 2430.0 kWh / 33950.00 BDT / peak 175.0 kWh

paste করার body: এই PDF-এর পৃষ্ঠা 21

17. L1 **LLM test** LLM test - প্রথম বাক্যটা অপ্ৰাসঙ্গিক, দ্বিতীয়টায় ঘরোয়া ভাষায় grid সীমা

- Rumour has it the canteen is closing for renovation next month. Meanwhile the substation crew asked us to be gentle on the grid between 6 and 10 PM - keep any single hour under 150 kWh.

প্রত্যাশিত: max_grid_window, hours [18, 19, 20, 21], max_grid_kwh 150.0

মোট: 2430.00 kWh / 33950.00 BDT / peak 175.00 kWh

paste করার body: এই PDF-এর পৃষ্ঠা 22

18. L2 **LLM test** LLM test - negation: রাতে চার্জ করা ঠিক আছে, ভোর ৪টা থেকে ৭টা নয়

- Charging through the night is fine. Just don't top the battery up between 4 and 7 in the morning while the inverter is being serviced.

প্রত্যাশিত: no_charge_window, hours [4, 5, 6]

মোট: 2430.00 kWh / 34010.00 BDT / peak 175.00 kWh

paste করার body: এই PDF-এর পৃষ্ঠা 23

19. M1 **multi-note** একসাথে দুইটা note - সোলার হ্রাস আর battery reserve

- Expect a 55% reduction in rooftop solar from 9 AM to 11 AM.
- Keep at least 115 kWh in the battery from 7 PM until 10 PM.

প্রত্যাশিত: solar_reduction [9, 10] factor 0.45, তারপর minimum_battery_reserve [19, 20, 21] 115.0

মোট: 2551.0 kWh / 36997.00 BDT / peak 200.0 kWh

paste করার body: এই PDF-এর পৃষ্ঠা 24

8. Error-handling case

ID	কেস	কী করবে	প্রত্যাশিত ফল
E1	POST-only endpoint-এ GET	ব্রাউজারে https://brian-sheffield-surely-fly.trycloudflare.com/optimize-energy খোলো	405 {"detail":"Method Not Allowed"}
E2	অজানা path	ব্রাউজারে https://brian-sheffield-surely-fly.trycloudflare.com/no-such-path খোলো	404 {"detail":"Not Found"}
E3	ফাঁকা request body	/docs → POST /optimize-energy → Try it out → body খালি করে দাও → Execute	422 {"error":{"type":"invalid_request",...}} - পরিষ্কার error, crash নেই, key ফাঁস নেই

সংখ্যায় শেষে কয়েকটা দশমিক বেশি দেখাতে পারে, যেমন 38365.000001, 38365.00-এর বদলে। এটা optimizer-এর tie-break tolerance; published optimum থেকে পার্থক্য ০.০০০০০১-এর বেশি নয়।

CASE SAMPLE-01 — SAMPLE-01 - official public sample case

পুরো ব্লকটা সিলেক্ট করে কপি করো, তারপর Swagger-এর Request body বাক্সে পেস্ট করো। ছবছ পেস্ট করো - একটাও অক্ষর বদলাবে না।

```
{
  "scenario_id": "SAMPLE-01",
  "operator_notes": [
    "Facilities will wash the rooftop solar panels from noon until 2 PM. During cleaning, usable solar should be treated as roughly 25% of the forecast.",
    "The sports office moved next month's registration deadline."
  ],
  "hours": [
    {"hour": 0, "demand_kwh": 90, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 1, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 2, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 3, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 4, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 5, "demand_kwh": 95, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 6, "demand_kwh": 110, "solar_kwh": 5, "tariff_bdt_per_kwh": 8},
    {"hour": 7, "demand_kwh": 130, "solar_kwh": 20, "tariff_bdt_per_kwh": 10},
    {"hour": 8, "demand_kwh": 150, "solar_kwh": 50, "tariff_bdt_per_kwh": 12},
    {"hour": 9, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 10, "demand_kwh": 175, "solar_kwh": 130, "tariff_bdt_per_kwh": 16},
    {"hour": 11, "demand_kwh": 180, "solar_kwh": 160, "tariff_bdt_per_kwh": 16},
    {"hour": 12, "demand_kwh": 185, "solar_kwh": 180, "tariff_bdt_per_kwh": 15},
    {"hour": 13, "demand_kwh": 180, "solar_kwh": 170, "tariff_bdt_per_kwh": 14},
    {"hour": 14, "demand_kwh": 170, "solar_kwh": 140, "tariff_bdt_per_kwh": 13},
    {"hour": 15, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 16, "demand_kwh": 170, "solar_kwh": 45, "tariff_bdt_per_kwh": 18},
    {"hour": 17, "demand_kwh": 185, "solar_kwh": 10, "tariff_bdt_per_kwh": 22},
    {"hour": 18, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 28},
    {"hour": 19, "demand_kwh": 215, "solar_kwh": 0, "tariff_bdt_per_kwh": 30},
    {"hour": 20, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 26},
    {"hour": 21, "demand_kwh": 175, "solar_kwh": 0, "tariff_bdt_per_kwh": 18},
    {"hour": 22, "demand_kwh": 135, "solar_kwh": 0, "tariff_bdt_per_kwh": 10},
    {"hour": 23, "demand_kwh": 105, "solar_kwh": 0, "tariff_bdt_per_kwh": 7}
  ],
  "battery": {
    "capacity_kwh": 220,
    "initial_energy_kwh": 110,
    "minimum_energy_kwh": 40,
    "max_charge_kwh_per_hour": 50,
    "max_discharge_kwh_per_hour": 50
  }
}
```

একই body এখানেও আছে <https://github.com/meherabmehu/GridWise-LLM-Energy-Optimizer/blob/main/docs/test-cases/request-bodies/sample-01.json>

CASE SAMPLE-02 — SAMPLE-02 - official public sample case

পুরো ব্লকটা সিলেক্ট করে কপি করো, তারপর Swagger-এর Request body বাক্সে পেস্ট করো। ছবছ পেস্ট করো - একটাও অক্ষর বদলাবে না।

```
{
  "scenario_id": "SAMPLE-02",
  "operator_notes": [
    "The battery charger will be isolated from 2 AM until 5 AM for electrical maintenance."
  ],
  "hours": [
    {"hour": 0, "demand_kwh": 100, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 1, "demand_kwh": 95, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 2, "demand_kwh": 90, "solar_kwh": 0, "tariff_bdt_per_kwh": 4},
    {"hour": 3, "demand_kwh": 90, "solar_kwh": 0, "tariff_bdt_per_kwh": 4},
    {"hour": 4, "demand_kwh": 95, "solar_kwh": 0, "tariff_bdt_per_kwh": 4},
    {"hour": 5, "demand_kwh": 105, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 6, "demand_kwh": 120, "solar_kwh": 0, "tariff_bdt_per_kwh": 7},
    {"hour": 7, "demand_kwh": 135, "solar_kwh": 10, "tariff_bdt_per_kwh": 9},
    {"hour": 8, "demand_kwh": 145, "solar_kwh": 30, "tariff_bdt_per_kwh": 11},
    {"hour": 9, "demand_kwh": 155, "solar_kwh": 55, "tariff_bdt_per_kwh": 13},
    {"hour": 10, "demand_kwh": 165, "solar_kwh": 80, "tariff_bdt_per_kwh": 15},
    {"hour": 11, "demand_kwh": 175, "solar_kwh": 100, "tariff_bdt_per_kwh": 16},
    {"hour": 12, "demand_kwh": 180, "solar_kwh": 110, "tariff_bdt_per_kwh": 16},
    {"hour": 13, "demand_kwh": 175, "solar_kwh": 105, "tariff_bdt_per_kwh": 15},
    {"hour": 14, "demand_kwh": 165, "solar_kwh": 85, "tariff_bdt_per_kwh": 14},
    {"hour": 15, "demand_kwh": 160, "solar_kwh": 60, "tariff_bdt_per_kwh": 15},
    {"hour": 16, "demand_kwh": 170, "solar_kwh": 30, "tariff_bdt_per_kwh": 19},
    {"hour": 17, "demand_kwh": 190, "solar_kwh": 10, "tariff_bdt_per_kwh": 24},
    {"hour": 18, "demand_kwh": 210, "solar_kwh": 0, "tariff_bdt_per_kwh": 31},
    {"hour": 19, "demand_kwh": 220, "solar_kwh": 0, "tariff_bdt_per_kwh": 33},
    {"hour": 20, "demand_kwh": 210, "solar_kwh": 0, "tariff_bdt_per_kwh": 29},
    {"hour": 21, "demand_kwh": 180, "solar_kwh": 0, "tariff_bdt_per_kwh": 20},
    {"hour": 22, "demand_kwh": 145, "solar_kwh": 0, "tariff_bdt_per_kwh": 11},
    {"hour": 23, "demand_kwh": 115, "solar_kwh": 0, "tariff_bdt_per_kwh": 7}
  ],
  "battery": {
    "capacity_kwh": 200,
    "initial_energy_kwh": 70,
    "minimum_energy_kwh": 30,
    "max_charge_kwh_per_hour": 55,
    "max_discharge_kwh_per_hour": 55
  }
}
```

একই body এখানেও আছে <https://github.com/meherabmehu/GridWise-LLM-Energy-Optimizer/blob/main/docs/test-cases/request-bodies/sample-02.json>

CASE SAMPLE-03 — SAMPLE-03 - official public sample case

পুরো ব্লকটা সিলেক্ট করে কপি করো, তারপর Swagger-এর Request body বাক্সে পেস্ট করো। ছবছ পেস্ট করো - একটাও অক্ষর বদলাবে না।

```
{
  "scenario_id": "SAMPLE-03",
  "operator_notes": [
    "Keep at least 50% of the battery capacity stored in the battery from 6 PM until 9 PM for emergency operations."
  ],
  "hours": [
    {"hour": 0, "demand_kwh": 90, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 1, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 2, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 3, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 4, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 5, "demand_kwh": 95, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 6, "demand_kwh": 110, "solar_kwh": 5, "tariff_bdt_per_kwh": 8},
    {"hour": 7, "demand_kwh": 130, "solar_kwh": 20, "tariff_bdt_per_kwh": 10},
    {"hour": 8, "demand_kwh": 150, "solar_kwh": 50, "tariff_bdt_per_kwh": 12},
    {"hour": 9, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 10, "demand_kwh": 175, "solar_kwh": 130, "tariff_bdt_per_kwh": 16},
    {"hour": 11, "demand_kwh": 180, "solar_kwh": 160, "tariff_bdt_per_kwh": 16},
    {"hour": 12, "demand_kwh": 185, "solar_kwh": 180, "tariff_bdt_per_kwh": 15},
    {"hour": 13, "demand_kwh": 180, "solar_kwh": 170, "tariff_bdt_per_kwh": 14},
    {"hour": 14, "demand_kwh": 170, "solar_kwh": 140, "tariff_bdt_per_kwh": 13},
    {"hour": 15, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 16, "demand_kwh": 170, "solar_kwh": 45, "tariff_bdt_per_kwh": 18},
    {"hour": 17, "demand_kwh": 185, "solar_kwh": 10, "tariff_bdt_per_kwh": 22},
    {"hour": 18, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 28},
    {"hour": 19, "demand_kwh": 215, "solar_kwh": 0, "tariff_bdt_per_kwh": 30},
    {"hour": 20, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 26},
    {"hour": 21, "demand_kwh": 175, "solar_kwh": 0, "tariff_bdt_per_kwh": 18},
    {"hour": 22, "demand_kwh": 135, "solar_kwh": 0, "tariff_bdt_per_kwh": 10},
    {"hour": 23, "demand_kwh": 105, "solar_kwh": 0, "tariff_bdt_per_kwh": 7}
  ],
  "battery": {
    "capacity_kwh": 200,
    "initial_energy_kwh": 120,
    "minimum_energy_kwh": 40,
    "max_charge_kwh_per_hour": 50,
    "max_discharge_kwh_per_hour": 50
  }
}
```

একই body এখানেও আছে <https://github.com/meherabmehu/GridWise-LLM-Energy-Optimizer/blob/main/docs/test-cases/request-bodies/sample-03.json>

CASE SAMPLE-04 — SAMPLE-04 - official public sample case

পুরো ব্লকটা সিলেক্ট করে কপি করো, তারপর Swagger-এর Request body বাক্সে পেস্ট করো। ছবছ পেস্ট করো - একটাও অক্ষর বদলাবে না।

```
{
  "scenario_id": "SAMPLE-04",
  "operator_notes": [
    "For protection testing, the battery must not discharge from 6 PM until 8 PM."
  ],
  "hours": [
    {"hour": 0, "demand_kwh": 95, "solar_kwh": 0, "tariff_bdt_per_kwh": 7},
    {"hour": 1, "demand_kwh": 90, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 2, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 3, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 4, "demand_kwh": 90, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 5, "demand_kwh": 100, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 6, "demand_kwh": 115, "solar_kwh": 5, "tariff_bdt_per_kwh": 8},
    {"hour": 7, "demand_kwh": 130, "solar_kwh": 15, "tariff_bdt_per_kwh": 10},
    {"hour": 8, "demand_kwh": 145, "solar_kwh": 40, "tariff_bdt_per_kwh": 12},
    {"hour": 9, "demand_kwh": 155, "solar_kwh": 75, "tariff_bdt_per_kwh": 14},
    {"hour": 10, "demand_kwh": 165, "solar_kwh": 110, "tariff_bdt_per_kwh": 16},
    {"hour": 11, "demand_kwh": 175, "solar_kwh": 145, "tariff_bdt_per_kwh": 16},
    {"hour": 12, "demand_kwh": 180, "solar_kwh": 165, "tariff_bdt_per_kwh": 15},
    {"hour": 13, "demand_kwh": 175, "solar_kwh": 155, "tariff_bdt_per_kwh": 14},
    {"hour": 14, "demand_kwh": 170, "solar_kwh": 125, "tariff_bdt_per_kwh": 13},
    {"hour": 15, "demand_kwh": 165, "solar_kwh": 80, "tariff_bdt_per_kwh": 14},
    {"hour": 16, "demand_kwh": 175, "solar_kwh": 35, "tariff_bdt_per_kwh": 17},
    {"hour": 17, "demand_kwh": 195, "solar_kwh": 5, "tariff_bdt_per_kwh": 21},
    {"hour": 18, "demand_kwh": 215, "solar_kwh": 0, "tariff_bdt_per_kwh": 29},
    {"hour": 19, "demand_kwh": 225, "solar_kwh": 0, "tariff_bdt_per_kwh": 32},
    {"hour": 20, "demand_kwh": 215, "solar_kwh": 0, "tariff_bdt_per_kwh": 30},
    {"hour": 21, "demand_kwh": 185, "solar_kwh": 0, "tariff_bdt_per_kwh": 20},
    {"hour": 22, "demand_kwh": 150, "solar_kwh": 0, "tariff_bdt_per_kwh": 11},
    {"hour": 23, "demand_kwh": 120, "solar_kwh": 0, "tariff_bdt_per_kwh": 8}
  ],
  "battery": {
    "capacity_kwh": 230,
    "initial_energy_kwh": 130,
    "minimum_energy_kwh": 40,
    "max_charge_kwh_per_hour": 55,
    "max_discharge_kwh_per_hour": 55
  }
}
```

একই body এখানেও আছে <https://github.com/meherabmehu/GridWise-LLM-Energy-Optimizer/blob/main/docs/test-cases/request-bodies/sample-04.json>

CASE SAMPLE-05 — SAMPLE-05 - official public sample case

পুরো ব্লকটা সিলেক্ট করে কপি করো, তারপর Swagger-এর Request body বাক্সে পেস্ট করো। ছবছ পেস্ট করো - একটাও অক্ষর বদলাবে না।

```
{
  "scenario_id": "SAMPLE-05",
  "operator_notes": [
    "From 6 PM until 9 PM, campus grid import must not exceed 155 kWh in any hour because the feeder is operating under a temporary limit."
  ],
  "hours": [
    {"hour": 0, "demand_kwh": 90, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 1, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 2, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 3, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 4, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 5, "demand_kwh": 95, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 6, "demand_kwh": 110, "solar_kwh": 5, "tariff_bdt_per_kwh": 8},
    {"hour": 7, "demand_kwh": 130, "solar_kwh": 20, "tariff_bdt_per_kwh": 10},
    {"hour": 8, "demand_kwh": 150, "solar_kwh": 50, "tariff_bdt_per_kwh": 12},
    {"hour": 9, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 10, "demand_kwh": 175, "solar_kwh": 130, "tariff_bdt_per_kwh": 16},
    {"hour": 11, "demand_kwh": 180, "solar_kwh": 160, "tariff_bdt_per_kwh": 16},
    {"hour": 12, "demand_kwh": 185, "solar_kwh": 180, "tariff_bdt_per_kwh": 15},
    {"hour": 13, "demand_kwh": 180, "solar_kwh": 170, "tariff_bdt_per_kwh": 14},
    {"hour": 14, "demand_kwh": 170, "solar_kwh": 140, "tariff_bdt_per_kwh": 13},
    {"hour": 15, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 16, "demand_kwh": 170, "solar_kwh": 45, "tariff_bdt_per_kwh": 18},
    {"hour": 17, "demand_kwh": 185, "solar_kwh": 10, "tariff_bdt_per_kwh": 22},
    {"hour": 18, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 28},
    {"hour": 19, "demand_kwh": 215, "solar_kwh": 0, "tariff_bdt_per_kwh": 30},
    {"hour": 20, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 26},
    {"hour": 21, "demand_kwh": 175, "solar_kwh": 0, "tariff_bdt_per_kwh": 18},
    {"hour": 22, "demand_kwh": 135, "solar_kwh": 0, "tariff_bdt_per_kwh": 10},
    {"hour": 23, "demand_kwh": 105, "solar_kwh": 0, "tariff_bdt_per_kwh": 7}
  ],
  "battery": {
    "capacity_kwh": 240,
    "initial_energy_kwh": 120,
    "minimum_energy_kwh": 30,
    "max_charge_kwh_per_hour": 60,
    "max_discharge_kwh_per_hour": 60
  }
}
```

একই body এখানেও আছে <https://github.com/meherabmehu/GridWise-LLM-Energy-Optimizer/blob/main/docs/test-cases/request-bodies/sample-05.json>

CASE SAMPLE-06 — SAMPLE-06 - official public sample case

পুরো ব্লকটা সিলেক্ট করে কপি করো, তারপর Swagger-এর Request body বাক্সে পেস্ট করো। ছবছ পেস্ট করো - একটাও অক্ষর বদলাবে না।

```
{
  "scenario_id": "SAMPLE-06",
  "operator_notes": [
    "Cloud cover during panel inspection will leave about half of the forecast solar output from 10 AM until noon.",
    "The charging circuit will be unavailable from 2 PM until 4 PM.",
    "The library is extending book-return hours next week."
  ],
  "hours": [
    {"hour": 0, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 1, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 2, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 3, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 4, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 5, "demand_kwh": 95, "solar_kwh": 0, "tariff_bdt_per_kwh": 7},
    {"hour": 6, "demand_kwh": 110, "solar_kwh": 5, "tariff_bdt_per_kwh": 8},
    {"hour": 7, "demand_kwh": 125, "solar_kwh": 20, "tariff_bdt_per_kwh": 9},
    {"hour": 8, "demand_kwh": 140, "solar_kwh": 55, "tariff_bdt_per_kwh": 11},
    {"hour": 9, "demand_kwh": 155, "solar_kwh": 100, "tariff_bdt_per_kwh": 13},
    {"hour": 10, "demand_kwh": 165, "solar_kwh": 150, "tariff_bdt_per_kwh": 15},
    {"hour": 11, "demand_kwh": 175, "solar_kwh": 190, "tariff_bdt_per_kwh": 16},
    {"hour": 12, "demand_kwh": 180, "solar_kwh": 210, "tariff_bdt_per_kwh": 16},
    {"hour": 13, "demand_kwh": 175, "solar_kwh": 200, "tariff_bdt_per_kwh": 15},
    {"hour": 14, "demand_kwh": 170, "solar_kwh": 160, "tariff_bdt_per_kwh": 14},
    {"hour": 15, "demand_kwh": 165, "solar_kwh": 100, "tariff_bdt_per_kwh": 15},
    {"hour": 16, "demand_kwh": 175, "solar_kwh": 50, "tariff_bdt_per_kwh": 18},
    {"hour": 17, "demand_kwh": 190, "solar_kwh": 15, "tariff_bdt_per_kwh": 22},
    {"hour": 18, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 27},
    {"hour": 19, "demand_kwh": 215, "solar_kwh": 0, "tariff_bdt_per_kwh": 29},
    {"hour": 20, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 27},
    {"hour": 21, "demand_kwh": 175, "solar_kwh": 0, "tariff_bdt_per_kwh": 18},
    {"hour": 22, "demand_kwh": 140, "solar_kwh": 0, "tariff_bdt_per_kwh": 10},
    {"hour": 23, "demand_kwh": 110, "solar_kwh": 0, "tariff_bdt_per_kwh": 7}
  ],
  "battery": {
    "capacity_kwh": 220,
    "initial_energy_kwh": 100,
    "minimum_energy_kwh": 35,
    "max_charge_kwh_per_hour": 50,
    "max_discharge_kwh_per_hour": 50
  }
}
```

একই body এখানেও আছে <https://github.com/meherabmehu/GridWise-LLM-Energy-Optimizer/blob/main/docs/test-cases/request-bodies/sample-06.json>

CASE SAMPLE-07 — SAMPLE-07 - official public sample case

পুরো ব্লকটা সিলেক্ট করে কপি করো, তারপর Swagger-এর Request body বাক্সে পেস্ট করো। ছবছ পেস্ট করো - একটাও অক্ষর বদলাবে না।

```
{
  "scenario_id": "SAMPLE-07",
  "operator_notes": [
    "Keep at least 90 kWh in the battery from 6 PM until 10 PM for emergency services.",
    "The evening transformer limit is 180 kWh of grid import from 7 PM until 9 PM."
  ],
  "hours": [
    {"hour": 0, "demand_kwh": 100, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 1, "demand_kwh": 95, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 2, "demand_kwh": 90, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 3, "demand_kwh": 90, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 4, "demand_kwh": 95, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 5, "demand_kwh": 105, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 6, "demand_kwh": 120, "solar_kwh": 5, "tariff_bdt_per_kwh": 8},
    {"hour": 7, "demand_kwh": 135, "solar_kwh": 20, "tariff_bdt_per_kwh": 10},
    {"hour": 8, "demand_kwh": 150, "solar_kwh": 50, "tariff_bdt_per_kwh": 12},
    {"hour": 9, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 10, "demand_kwh": 175, "solar_kwh": 135, "tariff_bdt_per_kwh": 15},
    {"hour": 11, "demand_kwh": 185, "solar_kwh": 170, "tariff_bdt_per_kwh": 16},
    {"hour": 12, "demand_kwh": 190, "solar_kwh": 190, "tariff_bdt_per_kwh": 16},
    {"hour": 13, "demand_kwh": 185, "solar_kwh": 180, "tariff_bdt_per_kwh": 15},
    {"hour": 14, "demand_kwh": 175, "solar_kwh": 145, "tariff_bdt_per_kwh": 14},
    {"hour": 15, "demand_kwh": 170, "solar_kwh": 95, "tariff_bdt_per_kwh": 15},
    {"hour": 16, "demand_kwh": 180, "solar_kwh": 45, "tariff_bdt_per_kwh": 18},
    {"hour": 17, "demand_kwh": 195, "solar_kwh": 10, "tariff_bdt_per_kwh": 23},
    {"hour": 18, "demand_kwh": 210, "solar_kwh": 0, "tariff_bdt_per_kwh": 29},
    {"hour": 19, "demand_kwh": 225, "solar_kwh": 0, "tariff_bdt_per_kwh": 32},
    {"hour": 20, "demand_kwh": 215, "solar_kwh": 0, "tariff_bdt_per_kwh": 30},
    {"hour": 21, "demand_kwh": 185, "solar_kwh": 0, "tariff_bdt_per_kwh": 21},
    {"hour": 22, "demand_kwh": 145, "solar_kwh": 0, "tariff_bdt_per_kwh": 11},
    {"hour": 23, "demand_kwh": 115, "solar_kwh": 0, "tariff_bdt_per_kwh": 7}
  ],
  "battery": {
    "capacity_kwh": 250,
    "initial_energy_kwh": 150,
    "minimum_energy_kwh": 40,
    "max_charge_kwh_per_hour": 60,
    "max_discharge_kwh_per_hour": 60
  }
}
```

একই body এখানেও আছে <https://github.com/meherabmehu/GridWise-LLM-Energy-Optimizer/blob/main/docs/test-cases/request-bodies/sample-07.json>

CASE SAMPLE-08 — SAMPLE-08 - official public sample case

পুরো ব্লকটা সিলেক্ট করে কপি করো, তারপর Swagger-এর Request body বাক্সে পেস্ট করো। ছবছ পেস্ট করো - একটাও অক্ষর বদলাবে না।

```
{
  "scenario_id": "SAMPLE-08",
  "operator_notes": [
    "Battery charging is disabled from 11 AM until 1 PM while technicians inspect the charger.",
    "Do not discharge the battery from 5 PM until 7 PM during relay testing."
  ],
  "hours": [
    {"hour": 0, "demand_kwh": 90, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 1, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 2, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 3, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 4, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 5, "demand_kwh": 95, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 6, "demand_kwh": 110, "solar_kwh": 0, "tariff_bdt_per_kwh": 8},
    {"hour": 7, "demand_kwh": 125, "solar_kwh": 15, "tariff_bdt_per_kwh": 10},
    {"hour": 8, "demand_kwh": 140, "solar_kwh": 40, "tariff_bdt_per_kwh": 12},
    {"hour": 9, "demand_kwh": 155, "solar_kwh": 80, "tariff_bdt_per_kwh": 14},
    {"hour": 10, "demand_kwh": 165, "solar_kwh": 120, "tariff_bdt_per_kwh": 15},
    {"hour": 11, "demand_kwh": 175, "solar_kwh": 155, "tariff_bdt_per_kwh": 16},
    {"hour": 12, "demand_kwh": 180, "solar_kwh": 175, "tariff_bdt_per_kwh": 15},
    {"hour": 13, "demand_kwh": 175, "solar_kwh": 165, "tariff_bdt_per_kwh": 14},
    {"hour": 14, "demand_kwh": 165, "solar_kwh": 130, "tariff_bdt_per_kwh": 13},
    {"hour": 15, "demand_kwh": 160, "solar_kwh": 85, "tariff_bdt_per_kwh": 14},
    {"hour": 16, "demand_kwh": 170, "solar_kwh": 40, "tariff_bdt_per_kwh": 18},
    {"hour": 17, "demand_kwh": 190, "solar_kwh": 10, "tariff_bdt_per_kwh": 24},
    {"hour": 18, "demand_kwh": 210, "solar_kwh": 0, "tariff_bdt_per_kwh": 30},
    {"hour": 19, "demand_kwh": 220, "solar_kwh": 0, "tariff_bdt_per_kwh": 31},
    {"hour": 20, "demand_kwh": 210, "solar_kwh": 0, "tariff_bdt_per_kwh": 28},
    {"hour": 21, "demand_kwh": 180, "solar_kwh": 0, "tariff_bdt_per_kwh": 19},
    {"hour": 22, "demand_kwh": 145, "solar_kwh": 0, "tariff_bdt_per_kwh": 10},
    {"hour": 23, "demand_kwh": 115, "solar_kwh": 0, "tariff_bdt_per_kwh": 7}
  ],
  "battery": {
    "capacity_kwh": 210,
    "initial_energy_kwh": 105,
    "minimum_energy_kwh": 35,
    "max_charge_kwh_per_hour": 50,
    "max_discharge_kwh_per_hour": 50
  }
}
```

একই body এখানেও আছে <https://github.com/meherabmehu/GridWise-LLM-Energy-Optimizer/blob/main/docs/test-cases/request-bodies/sample-08.json>

CASE SAMPLE-09 — SAMPLE-09 - official public sample case

পুরো ব্লকটা সিলেক্ট করে কপি করো, তারপর Swagger-এর Request body বাক্সে পেস্ট করো। ছবছ পেস্ট করো - একটাও অক্ষর বদলাবে না।

```
{
  "scenario_id": "SAMPLE-09",
  "operator_notes": [
    "Expect an 80% reduction in rooftop solar between 11 AM and 2 PM because of inverter work.",
    "The student affairs office will publish club notices tomorrow."
  ],
  "hours": [
    {"hour": 0, "demand_kwh": 90, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 1, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 2, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 3, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 4, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 5, "demand_kwh": 95, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 6, "demand_kwh": 105, "solar_kwh": 5, "tariff_bdt_per_kwh": 8},
    {"hour": 7, "demand_kwh": 120, "solar_kwh": 25, "tariff_bdt_per_kwh": 10},
    {"hour": 8, "demand_kwh": 135, "solar_kwh": 65, "tariff_bdt_per_kwh": 12},
    {"hour": 9, "demand_kwh": 150, "solar_kwh": 120, "tariff_bdt_per_kwh": 13},
    {"hour": 10, "demand_kwh": 165, "solar_kwh": 180, "tariff_bdt_per_kwh": 14},
    {"hour": 11, "demand_kwh": 175, "solar_kwh": 230, "tariff_bdt_per_kwh": 15},
    {"hour": 12, "demand_kwh": 180, "solar_kwh": 260, "tariff_bdt_per_kwh": 15},
    {"hour": 13, "demand_kwh": 175, "solar_kwh": 240, "tariff_bdt_per_kwh": 14},
    {"hour": 14, "demand_kwh": 165, "solar_kwh": 190, "tariff_bdt_per_kwh": 13},
    {"hour": 15, "demand_kwh": 160, "solar_kwh": 120, "tariff_bdt_per_kwh": 14},
    {"hour": 16, "demand_kwh": 170, "solar_kwh": 55, "tariff_bdt_per_kwh": 18},
    {"hour": 17, "demand_kwh": 185, "solar_kwh": 10, "tariff_bdt_per_kwh": 22},
    {"hour": 18, "demand_kwh": 200, "solar_kwh": 0, "tariff_bdt_per_kwh": 27},
    {"hour": 19, "demand_kwh": 210, "solar_kwh": 0, "tariff_bdt_per_kwh": 29},
    {"hour": 20, "demand_kwh": 200, "solar_kwh": 0, "tariff_bdt_per_kwh": 26},
    {"hour": 21, "demand_kwh": 170, "solar_kwh": 0, "tariff_bdt_per_kwh": 18},
    {"hour": 22, "demand_kwh": 135, "solar_kwh": 0, "tariff_bdt_per_kwh": 10},
    {"hour": 23, "demand_kwh": 105, "solar_kwh": 0, "tariff_bdt_per_kwh": 7}
  ],
  "battery": {
    "capacity_kwh": 240,
    "initial_energy_kwh": 120,
    "minimum_energy_kwh": 40,
    "max_charge_kwh_per_hour": 60,
    "max_discharge_kwh_per_hour": 60
  }
}
```

একই body এখানেও আছে <https://github.com/meherabmehu/GridWise-LLM-Energy-Optimizer/blob/main/docs/test-cases/request-bodies/sample-09.json>

CASE SAMPLE-10 — SAMPLE-10 - official public sample case

পুরো ব্লকটা সিলেক্ট করে কপি করো, তারপর Swagger-এর Request body বাক্সে পেস্ট করো। ছবছ পেস্ট করো - একটাও অক্ষর বদলাবে না।

```
{
  "scenario_id": "SAMPLE-10",
  "operator_notes": [
    "The data center requires at least 80 kWh to remain in the battery from 6 PM until 10 PM.",
    "Grid intake must stay at or below 190 kWh from 7 PM until 10 PM while the substation is constrained.",
    "A seminar room booking was moved to next week."
  ],
  "hours": [
    {"hour": 0, "demand_kwh": 105, "solar_kwh": 0, "tariff_bdt_per_kwh": 7},
    {"hour": 1, "demand_kwh": 100, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 2, "demand_kwh": 95, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 3, "demand_kwh": 95, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 4, "demand_kwh": 100, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 5, "demand_kwh": 110, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 6, "demand_kwh": 125, "solar_kwh": 5, "tariff_bdt_per_kwh": 8},
    {"hour": 7, "demand_kwh": 140, "solar_kwh": 20, "tariff_bdt_per_kwh": 10},
    {"hour": 8, "demand_kwh": 155, "solar_kwh": 50, "tariff_bdt_per_kwh": 12},
    {"hour": 9, "demand_kwh": 170, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 10, "demand_kwh": 180, "solar_kwh": 130, "tariff_bdt_per_kwh": 16},
    {"hour": 11, "demand_kwh": 190, "solar_kwh": 165, "tariff_bdt_per_kwh": 17},
    {"hour": 12, "demand_kwh": 195, "solar_kwh": 185, "tariff_bdt_per_kwh": 16},
    {"hour": 13, "demand_kwh": 190, "solar_kwh": 175, "tariff_bdt_per_kwh": 15},
    {"hour": 14, "demand_kwh": 180, "solar_kwh": 140, "tariff_bdt_per_kwh": 14},
    {"hour": 15, "demand_kwh": 175, "solar_kwh": 90, "tariff_bdt_per_kwh": 15},
    {"hour": 16, "demand_kwh": 185, "solar_kwh": 40, "tariff_bdt_per_kwh": 19},
    {"hour": 17, "demand_kwh": 200, "solar_kwh": 10, "tariff_bdt_per_kwh": 24},
    {"hour": 18, "demand_kwh": 215, "solar_kwh": 0, "tariff_bdt_per_kwh": 30},
    {"hour": 19, "demand_kwh": 230, "solar_kwh": 0, "tariff_bdt_per_kwh": 34},
    {"hour": 20, "demand_kwh": 220, "solar_kwh": 0, "tariff_bdt_per_kwh": 31},
    {"hour": 21, "demand_kwh": 190, "solar_kwh": 0, "tariff_bdt_per_kwh": 21},
    {"hour": 22, "demand_kwh": 150, "solar_kwh": 0, "tariff_bdt_per_kwh": 11},
    {"hour": 23, "demand_kwh": 120, "solar_kwh": 0, "tariff_bdt_per_kwh": 8}
  ],
  "battery": {
    "capacity_kwh": 260,
    "initial_energy_kwh": 140,
    "minimum_energy_kwh": 40,
    "max_charge_kwh_per_hour": 65,
    "max_discharge_kwh_per_hour": 65
  }
}
```

একই body এখানেও আছে <https://github.com/meherabmehu/GridWise-LLM-Energy-Optimizer/blob/main/docs/test-cases/request-bodies/sample-10.json>

CASE P1 — solar_reduction - ৫৫% হ্রাস, সকাল ৯টা থেকে ১১টা

পুরো ব্লকটা সিলেক্ট করে কপি করো, তারপর Swagger-এর Request body বাক্সে পেস্ট করো। ছবছ পেস্ট করো - একটাও অক্ষর বদলাবে না।

```
{
  "scenario_id": "PARAPHRASE-P1",
  "operator_notes": [
    "Expect a 55% reduction in rooftop solar from 9 AM to 11 AM."
  ],
  "hours": [
    {"hour": 0, "demand_kwh": 90, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 1, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 2, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 3, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 4, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 5, "demand_kwh": 95, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 6, "demand_kwh": 110, "solar_kwh": 5, "tariff_bdt_per_kwh": 8},
    {"hour": 7, "demand_kwh": 130, "solar_kwh": 20, "tariff_bdt_per_kwh": 10},
    {"hour": 8, "demand_kwh": 150, "solar_kwh": 50, "tariff_bdt_per_kwh": 12},
    {"hour": 9, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 10, "demand_kwh": 175, "solar_kwh": 130, "tariff_bdt_per_kwh": 16},
    {"hour": 11, "demand_kwh": 180, "solar_kwh": 160, "tariff_bdt_per_kwh": 16},
    {"hour": 12, "demand_kwh": 185, "solar_kwh": 180, "tariff_bdt_per_kwh": 15},
    {"hour": 13, "demand_kwh": 180, "solar_kwh": 170, "tariff_bdt_per_kwh": 14},
    {"hour": 14, "demand_kwh": 170, "solar_kwh": 140, "tariff_bdt_per_kwh": 13},
    {"hour": 15, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 16, "demand_kwh": 170, "solar_kwh": 45, "tariff_bdt_per_kwh": 18},
    {"hour": 17, "demand_kwh": 185, "solar_kwh": 10, "tariff_bdt_per_kwh": 22},
    {"hour": 18, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 28},
    {"hour": 19, "demand_kwh": 215, "solar_kwh": 0, "tariff_bdt_per_kwh": 30},
    {"hour": 20, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 26},
    {"hour": 21, "demand_kwh": 175, "solar_kwh": 0, "tariff_bdt_per_kwh": 18},
    {"hour": 22, "demand_kwh": 135, "solar_kwh": 0, "tariff_bdt_per_kwh": 10},
    {"hour": 23, "demand_kwh": 105, "solar_kwh": 0, "tariff_bdt_per_kwh": 7}
  ],
  "battery": {
    "capacity_kwh": 240,
    "initial_energy_kwh": 120,
    "minimum_energy_kwh": 30,
    "max_charge_kwh_per_hour": 60,
    "max_discharge_kwh_per_hour": 60
  }
}
```

একই body এখানেও আছে <https://github.com/meherabmehu/GridWise-LLM-Energy-Optimizer/blob/main/docs/test-cases/request-bodies/p1.json>

CASE P2 — minimum_battery_reserve - অন্তত ১১৫ kWh, সন্ধ্যা ৭টা থেকে ১০টা

পুরো ব্লকটা সিলেক্ট করে কপি করো, তারপর Swagger-এর Request body বাক্সে পেস্ট করো। ছবছ পেস্ট করো - একটাও অক্ষর বদলাবে না।

```
{
  "scenario_id": "PARAPHRASE-P2",
  "operator_notes": [
    "Keep at least 115 kWh in the battery from 7 PM until 10 PM."
  ],
  "hours": [
    {"hour": 0, "demand_kwh": 90, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 1, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 2, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 3, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 4, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 5, "demand_kwh": 95, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 6, "demand_kwh": 110, "solar_kwh": 5, "tariff_bdt_per_kwh": 8},
    {"hour": 7, "demand_kwh": 130, "solar_kwh": 20, "tariff_bdt_per_kwh": 10},
    {"hour": 8, "demand_kwh": 150, "solar_kwh": 50, "tariff_bdt_per_kwh": 12},
    {"hour": 9, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 10, "demand_kwh": 175, "solar_kwh": 130, "tariff_bdt_per_kwh": 16},
    {"hour": 11, "demand_kwh": 180, "solar_kwh": 160, "tariff_bdt_per_kwh": 16},
    {"hour": 12, "demand_kwh": 185, "solar_kwh": 180, "tariff_bdt_per_kwh": 15},
    {"hour": 13, "demand_kwh": 180, "solar_kwh": 170, "tariff_bdt_per_kwh": 14},
    {"hour": 14, "demand_kwh": 170, "solar_kwh": 140, "tariff_bdt_per_kwh": 13},
    {"hour": 15, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 16, "demand_kwh": 170, "solar_kwh": 45, "tariff_bdt_per_kwh": 18},
    {"hour": 17, "demand_kwh": 185, "solar_kwh": 10, "tariff_bdt_per_kwh": 22},
    {"hour": 18, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 28},
    {"hour": 19, "demand_kwh": 215, "solar_kwh": 0, "tariff_bdt_per_kwh": 30},
    {"hour": 20, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 26},
    {"hour": 21, "demand_kwh": 175, "solar_kwh": 0, "tariff_bdt_per_kwh": 18},
    {"hour": 22, "demand_kwh": 135, "solar_kwh": 0, "tariff_bdt_per_kwh": 10},
    {"hour": 23, "demand_kwh": 105, "solar_kwh": 0, "tariff_bdt_per_kwh": 7}
  ],
  "battery": {
    "capacity_kwh": 240,
    "initial_energy_kwh": 120,
    "minimum_energy_kwh": 30,
    "max_charge_kwh_per_hour": 60,
    "max_discharge_kwh_per_hour": 60
  }
}
```

একই body এখানেও আছে <https://github.com/meherabmehu/GridWise-LLM-Energy-Optimizer/blob/main/docs/test-cases/request-bodies/p2.json>

CASE P3 — no_charge_window - চার্জ ক্লা, রাত ২টা থেকে ৫টা

পুরো ব্লকটা সিলেক্ট করে কপি করো, তারপর Swagger-এর Request body বাক্সে পেস্ট করো। ছবছ পেস্ট করো - একটাও অক্ষর বদলাবে না।

```
{
  "scenario_id": "PARAPHRASE-P3",
  "operator_notes": [
    "Please don't charge the battery between 2 AM and 5 AM."
  ],
  "hours": [
    {"hour": 0, "demand_kwh": 90, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 1, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 2, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 3, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 4, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 5, "demand_kwh": 95, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 6, "demand_kwh": 110, "solar_kwh": 5, "tariff_bdt_per_kwh": 8},
    {"hour": 7, "demand_kwh": 130, "solar_kwh": 20, "tariff_bdt_per_kwh": 10},
    {"hour": 8, "demand_kwh": 150, "solar_kwh": 50, "tariff_bdt_per_kwh": 12},
    {"hour": 9, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 10, "demand_kwh": 175, "solar_kwh": 130, "tariff_bdt_per_kwh": 16},
    {"hour": 11, "demand_kwh": 180, "solar_kwh": 160, "tariff_bdt_per_kwh": 16},
    {"hour": 12, "demand_kwh": 185, "solar_kwh": 180, "tariff_bdt_per_kwh": 15},
    {"hour": 13, "demand_kwh": 180, "solar_kwh": 170, "tariff_bdt_per_kwh": 14},
    {"hour": 14, "demand_kwh": 170, "solar_kwh": 140, "tariff_bdt_per_kwh": 13},
    {"hour": 15, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 16, "demand_kwh": 170, "solar_kwh": 45, "tariff_bdt_per_kwh": 18},
    {"hour": 17, "demand_kwh": 185, "solar_kwh": 10, "tariff_bdt_per_kwh": 22},
    {"hour": 18, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 28},
    {"hour": 19, "demand_kwh": 215, "solar_kwh": 0, "tariff_bdt_per_kwh": 30},
    {"hour": 20, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 26},
    {"hour": 21, "demand_kwh": 175, "solar_kwh": 0, "tariff_bdt_per_kwh": 18},
    {"hour": 22, "demand_kwh": 135, "solar_kwh": 0, "tariff_bdt_per_kwh": 10},
    {"hour": 23, "demand_kwh": 105, "solar_kwh": 0, "tariff_bdt_per_kwh": 7}
  ],
  "battery": {
    "capacity_kwh": 240,
    "initial_energy_kwh": 120,
    "minimum_energy_kwh": 30,
    "max_charge_kwh_per_hour": 60,
    "max_discharge_kwh_per_hour": 60
  }
}
```

একই body এখানেও আছে <https://github.com/meherabmehu/GridWise-LLM-Energy-Optimizer/blob/main/docs/test-cases/request-bodies/p3.json>

CASE P4 — no_discharge_window - ডিসচার্জ কক্ষ, বিকাল ৪টা থেকে ৬টা

পুরো ব্লকটা সিলেক্ট করে কপি করো, তারপর Swagger-এর Request body বাক্সে পেস্ট করো। ছবছ পেস্ট করো - একটাও অক্ষর বদলাবে না।

```
{
  "scenario_id": "PARAPHRASE-P4",
  "operator_notes": [
    "Battery discharge is prohibited between 4 PM and 6 PM during relay work."
  ],
  "hours": [
    {"hour": 0, "demand_kwh": 90, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 1, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 2, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 3, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 4, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 5, "demand_kwh": 95, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 6, "demand_kwh": 110, "solar_kwh": 5, "tariff_bdt_per_kwh": 8},
    {"hour": 7, "demand_kwh": 130, "solar_kwh": 20, "tariff_bdt_per_kwh": 10},
    {"hour": 8, "demand_kwh": 150, "solar_kwh": 50, "tariff_bdt_per_kwh": 12},
    {"hour": 9, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 10, "demand_kwh": 175, "solar_kwh": 130, "tariff_bdt_per_kwh": 16},
    {"hour": 11, "demand_kwh": 180, "solar_kwh": 160, "tariff_bdt_per_kwh": 16},
    {"hour": 12, "demand_kwh": 185, "solar_kwh": 180, "tariff_bdt_per_kwh": 15},
    {"hour": 13, "demand_kwh": 180, "solar_kwh": 170, "tariff_bdt_per_kwh": 14},
    {"hour": 14, "demand_kwh": 170, "solar_kwh": 140, "tariff_bdt_per_kwh": 13},
    {"hour": 15, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 16, "demand_kwh": 170, "solar_kwh": 45, "tariff_bdt_per_kwh": 18},
    {"hour": 17, "demand_kwh": 185, "solar_kwh": 10, "tariff_bdt_per_kwh": 22},
    {"hour": 18, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 28},
    {"hour": 19, "demand_kwh": 215, "solar_kwh": 0, "tariff_bdt_per_kwh": 30},
    {"hour": 20, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 26},
    {"hour": 21, "demand_kwh": 175, "solar_kwh": 0, "tariff_bdt_per_kwh": 18},
    {"hour": 22, "demand_kwh": 135, "solar_kwh": 0, "tariff_bdt_per_kwh": 10},
    {"hour": 23, "demand_kwh": 105, "solar_kwh": 0, "tariff_bdt_per_kwh": 7}
  ],
  "battery": {
    "capacity_kwh": 240,
    "initial_energy_kwh": 120,
    "minimum_energy_kwh": 30,
    "max_charge_kwh_per_hour": 60,
    "max_discharge_kwh_per_hour": 60
  }
}
```

একই body এখানেও আছে <https://github.com/meherabmehu/GridWise-LLM-Energy-Optimizer/blob/main/docs/test-cases/request-bodies/p4.json>

CASE P5 — max_grid_window - import সর্বোচ্চ ১৫৫ kWh, সন্ধ্যা ৬টা থেকে ৯টা

পুরো ব্লকটা সিলেক্ট করে কপি করো, তারপর Swagger-এর Request body বাক্সে পেস্ট করো। ছবছ পেস্ট করো - একটাও অক্ষর বদলাবে না।

```
{
  "scenario_id": "PARAPHRASE-P5",
  "operator_notes": [
    "Between 6 PM and 9 PM the campus feeder is limited, so keep hourly grid import under 155 kWh."
  ],
  "hours": [
    {"hour": 0, "demand_kwh": 90, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 1, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 2, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 3, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 4, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 5, "demand_kwh": 95, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 6, "demand_kwh": 110, "solar_kwh": 5, "tariff_bdt_per_kwh": 8},
    {"hour": 7, "demand_kwh": 130, "solar_kwh": 20, "tariff_bdt_per_kwh": 10},
    {"hour": 8, "demand_kwh": 150, "solar_kwh": 50, "tariff_bdt_per_kwh": 12},
    {"hour": 9, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 10, "demand_kwh": 175, "solar_kwh": 130, "tariff_bdt_per_kwh": 16},
    {"hour": 11, "demand_kwh": 180, "solar_kwh": 160, "tariff_bdt_per_kwh": 16},
    {"hour": 12, "demand_kwh": 185, "solar_kwh": 180, "tariff_bdt_per_kwh": 15},
    {"hour": 13, "demand_kwh": 180, "solar_kwh": 170, "tariff_bdt_per_kwh": 14},
    {"hour": 14, "demand_kwh": 170, "solar_kwh": 140, "tariff_bdt_per_kwh": 13},
    {"hour": 15, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 16, "demand_kwh": 170, "solar_kwh": 45, "tariff_bdt_per_kwh": 18},
    {"hour": 17, "demand_kwh": 185, "solar_kwh": 10, "tariff_bdt_per_kwh": 22},
    {"hour": 18, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 28},
    {"hour": 19, "demand_kwh": 215, "solar_kwh": 0, "tariff_bdt_per_kwh": 30},
    {"hour": 20, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 26},
    {"hour": 21, "demand_kwh": 175, "solar_kwh": 0, "tariff_bdt_per_kwh": 18},
    {"hour": 22, "demand_kwh": 135, "solar_kwh": 0, "tariff_bdt_per_kwh": 10},
    {"hour": 23, "demand_kwh": 105, "solar_kwh": 0, "tariff_bdt_per_kwh": 7}
  ],
  "battery": {
    "capacity_kwh": 240,
    "initial_energy_kwh": 120,
    "minimum_energy_kwh": 30,
    "max_charge_kwh_per_hour": 60,
    "max_discharge_kwh_per_hour": 60
  }
}
```

একই body এখানেও আছে <https://github.com/meherabmehu/GridWise-LLM-Energy-Optimizer/blob/main/docs/test-cases/request-bodies/p5.json>

CASE P6 — no_op - যে note বাদ দিতে হবে

পুরো ব্লকটা সিলেক্ট করে কপি করো, তারপর Swagger-এর Request body বাক্সে পেস্ট করো। ছবছ পেস্ট করো - একটাও অক্ষর বদলাবে না।

```
{
  "scenario_id": "PARAPHRASE-P6",
  "operator_notes": [
    "The library extended its opening hours during exam week."
  ],
  "hours": [
    {"hour": 0, "demand_kwh": 90, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 1, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 2, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 3, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 4, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 5, "demand_kwh": 95, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 6, "demand_kwh": 110, "solar_kwh": 5, "tariff_bdt_per_kwh": 8},
    {"hour": 7, "demand_kwh": 130, "solar_kwh": 20, "tariff_bdt_per_kwh": 10},
    {"hour": 8, "demand_kwh": 150, "solar_kwh": 50, "tariff_bdt_per_kwh": 12},
    {"hour": 9, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 10, "demand_kwh": 175, "solar_kwh": 130, "tariff_bdt_per_kwh": 16},
    {"hour": 11, "demand_kwh": 180, "solar_kwh": 160, "tariff_bdt_per_kwh": 16},
    {"hour": 12, "demand_kwh": 185, "solar_kwh": 180, "tariff_bdt_per_kwh": 15},
    {"hour": 13, "demand_kwh": 180, "solar_kwh": 170, "tariff_bdt_per_kwh": 14},
    {"hour": 14, "demand_kwh": 170, "solar_kwh": 140, "tariff_bdt_per_kwh": 13},
    {"hour": 15, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 16, "demand_kwh": 170, "solar_kwh": 45, "tariff_bdt_per_kwh": 18},
    {"hour": 17, "demand_kwh": 185, "solar_kwh": 10, "tariff_bdt_per_kwh": 22},
    {"hour": 18, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 28},
    {"hour": 19, "demand_kwh": 215, "solar_kwh": 0, "tariff_bdt_per_kwh": 30},
    {"hour": 20, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 26},
    {"hour": 21, "demand_kwh": 175, "solar_kwh": 0, "tariff_bdt_per_kwh": 18},
    {"hour": 22, "demand_kwh": 135, "solar_kwh": 0, "tariff_bdt_per_kwh": 10},
    {"hour": 23, "demand_kwh": 105, "solar_kwh": 0, "tariff_bdt_per_kwh": 7}
  ],
  "battery": {
    "capacity_kwh": 240,
    "initial_energy_kwh": 120,
    "minimum_energy_kwh": 30,
    "max_charge_kwh_per_hour": 60,
    "max_discharge_kwh_per_hour": 60
  }
}
```

একই body এখানেও আছে <https://github.com/meherabmehu/GridWise-LLM-Energy-Optimizer/blob/main/docs/test-cases/request-bodies/p6.json>

CASE L1 — LLM test - প্রথম বাক্যটা অপ্রাসঙ্গিক, দ্বিতীয়টায় ঘরোয়া ভাষায় grid সীমা

পুরো ব্লকটা সিলেক্ট করে কপি করো, তারপর Swagger-এর Request body বাক্সে পেস্ট করো। ছবছ পেস্ট করো - একটাও অক্ষর বদলাবে না।

```
{
  "scenario_id": "LLM-TEST-1",
  "operator_notes": [
    "Rumour has it the canteen is closing for renovation next month. Meanwhile the substation crew asked us to be gentle on the grid between 6 and 10 PM - keep any single hour under 150 kwh",
  ],
  "hours": [
    {"hour": 0, "demand_kwh": 90, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 1, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 2, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 3, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 4, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 5, "demand_kwh": 95, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 6, "demand_kwh": 110, "solar_kwh": 5, "tariff_bdt_per_kwh": 8},
    {"hour": 7, "demand_kwh": 130, "solar_kwh": 20, "tariff_bdt_per_kwh": 10},
    {"hour": 8, "demand_kwh": 150, "solar_kwh": 50, "tariff_bdt_per_kwh": 12},
    {"hour": 9, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 10, "demand_kwh": 175, "solar_kwh": 130, "tariff_bdt_per_kwh": 16},
    {"hour": 11, "demand_kwh": 180, "solar_kwh": 160, "tariff_bdt_per_kwh": 16},
    {"hour": 12, "demand_kwh": 185, "solar_kwh": 180, "tariff_bdt_per_kwh": 15},
    {"hour": 13, "demand_kwh": 180, "solar_kwh": 170, "tariff_bdt_per_kwh": 14},
    {"hour": 14, "demand_kwh": 170, "solar_kwh": 140, "tariff_bdt_per_kwh": 13},
    {"hour": 15, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 16, "demand_kwh": 170, "solar_kwh": 45, "tariff_bdt_per_kwh": 18},
    {"hour": 17, "demand_kwh": 185, "solar_kwh": 10, "tariff_bdt_per_kwh": 22},
    {"hour": 18, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 28},
    {"hour": 19, "demand_kwh": 215, "solar_kwh": 0, "tariff_bdt_per_kwh": 30},
    {"hour": 20, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 26},
    {"hour": 21, "demand_kwh": 175, "solar_kwh": 0, "tariff_bdt_per_kwh": 18},
    {"hour": 22, "demand_kwh": 135, "solar_kwh": 0, "tariff_bdt_per_kwh": 10},
    {"hour": 23, "demand_kwh": 105, "solar_kwh": 0, "tariff_bdt_per_kwh": 7}
  ],
  "battery": {
    "capacity_kwh": 240,
    "initial_energy_kwh": 120,
    "minimum_energy_kwh": 30,
    "max_charge_kwh_per_hour": 60,
    "max_discharge_kwh_per_hour": 60
  }
}
```

একই body এখানেও আছে <https://github.com/meherabmehu/GridWise-LLM-Energy-Optimizer/blob/main/docs/test-cases/request-bodies/l1.json>

CASE L2 — LLM test - negation: রাতে চার্জ করা ঠিক আছে, ভোর ৪টা থেকে ৭টা নয়

পুরো ব্লকটা সিলেক্ট করে কপি করো, তারপর Swagger-এর Request body বাক্সে পেস্ট করো। ছবছ পেস্ট করো - একটাও অক্ষর বদলাবে না।

```
{
  "scenario_id": "LLM-TEST-2",
  "operator_notes": [
    "Charging through the night is fine. Just don't top the battery up between 4 and 7 in the morning while the inverter is being serviced."
  ],
  "hours": [
    {"hour": 0, "demand_kwh": 90, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 1, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 2, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 3, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 4, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 5, "demand_kwh": 95, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 6, "demand_kwh": 110, "solar_kwh": 5, "tariff_bdt_per_kwh": 8},
    {"hour": 7, "demand_kwh": 130, "solar_kwh": 20, "tariff_bdt_per_kwh": 10},
    {"hour": 8, "demand_kwh": 150, "solar_kwh": 50, "tariff_bdt_per_kwh": 12},
    {"hour": 9, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 10, "demand_kwh": 175, "solar_kwh": 130, "tariff_bdt_per_kwh": 16},
    {"hour": 11, "demand_kwh": 180, "solar_kwh": 160, "tariff_bdt_per_kwh": 16},
    {"hour": 12, "demand_kwh": 185, "solar_kwh": 180, "tariff_bdt_per_kwh": 15},
    {"hour": 13, "demand_kwh": 180, "solar_kwh": 170, "tariff_bdt_per_kwh": 14},
    {"hour": 14, "demand_kwh": 170, "solar_kwh": 140, "tariff_bdt_per_kwh": 13},
    {"hour": 15, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 16, "demand_kwh": 170, "solar_kwh": 45, "tariff_bdt_per_kwh": 18},
    {"hour": 17, "demand_kwh": 185, "solar_kwh": 10, "tariff_bdt_per_kwh": 22},
    {"hour": 18, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 28},
    {"hour": 19, "demand_kwh": 215, "solar_kwh": 0, "tariff_bdt_per_kwh": 30},
    {"hour": 20, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 26},
    {"hour": 21, "demand_kwh": 175, "solar_kwh": 0, "tariff_bdt_per_kwh": 18},
    {"hour": 22, "demand_kwh": 135, "solar_kwh": 0, "tariff_bdt_per_kwh": 10},
    {"hour": 23, "demand_kwh": 105, "solar_kwh": 0, "tariff_bdt_per_kwh": 7}
  ],
  "battery": {
    "capacity_kwh": 240,
    "initial_energy_kwh": 120,
    "minimum_energy_kwh": 30,
    "max_charge_kwh_per_hour": 60,
    "max_discharge_kwh_per_hour": 60
  }
}
```

একই body এখানেও আছে <https://github.com/meherabmehu/GridWise-LLM-Energy-Optimizer/blob/main/docs/test-cases/request-bodies/l2.json>

CASE M1 — একসাথে দুইটা note - সোলার হ্রাস আর battery reserve

পুরো ব্লকটা সিলেক্ট করে কপি করো, তারপর Swagger-এর Request body বাক্সে পেস্ট করো। ছবছ পেস্ট করো - একটাও অক্ষর বদলাবে না।

```
{
  "scenario_id": "MULTI-1",
  "operator_notes": [
    "Expect a 55% reduction in rooftop solar from 9 AM to 11 AM.",
    "Keep at least 115 kWh in the battery from 7 PM until 10 PM."
  ],
  "hours": [
    {"hour": 0, "demand_kwh": 90, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 1, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 2, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 3, "demand_kwh": 80, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 4, "demand_kwh": 85, "solar_kwh": 0, "tariff_bdt_per_kwh": 5},
    {"hour": 5, "demand_kwh": 95, "solar_kwh": 0, "tariff_bdt_per_kwh": 6},
    {"hour": 6, "demand_kwh": 110, "solar_kwh": 5, "tariff_bdt_per_kwh": 8},
    {"hour": 7, "demand_kwh": 130, "solar_kwh": 20, "tariff_bdt_per_kwh": 10},
    {"hour": 8, "demand_kwh": 150, "solar_kwh": 50, "tariff_bdt_per_kwh": 12},
    {"hour": 9, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 10, "demand_kwh": 175, "solar_kwh": 130, "tariff_bdt_per_kwh": 16},
    {"hour": 11, "demand_kwh": 180, "solar_kwh": 160, "tariff_bdt_per_kwh": 16},
    {"hour": 12, "demand_kwh": 185, "solar_kwh": 180, "tariff_bdt_per_kwh": 15},
    {"hour": 13, "demand_kwh": 180, "solar_kwh": 170, "tariff_bdt_per_kwh": 14},
    {"hour": 14, "demand_kwh": 170, "solar_kwh": 140, "tariff_bdt_per_kwh": 13},
    {"hour": 15, "demand_kwh": 165, "solar_kwh": 90, "tariff_bdt_per_kwh": 14},
    {"hour": 16, "demand_kwh": 170, "solar_kwh": 45, "tariff_bdt_per_kwh": 18},
    {"hour": 17, "demand_kwh": 185, "solar_kwh": 10, "tariff_bdt_per_kwh": 22},
    {"hour": 18, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 28},
    {"hour": 19, "demand_kwh": 215, "solar_kwh": 0, "tariff_bdt_per_kwh": 30},
    {"hour": 20, "demand_kwh": 205, "solar_kwh": 0, "tariff_bdt_per_kwh": 26},
    {"hour": 21, "demand_kwh": 175, "solar_kwh": 0, "tariff_bdt_per_kwh": 18},
    {"hour": 22, "demand_kwh": 135, "solar_kwh": 0, "tariff_bdt_per_kwh": 10},
    {"hour": 23, "demand_kwh": 105, "solar_kwh": 0, "tariff_bdt_per_kwh": 7}
  ],
  "battery": {
    "capacity_kwh": 240,
    "initial_energy_kwh": 120,
    "minimum_energy_kwh": 30,
    "max_charge_kwh_per_hour": 60,
    "max_discharge_kwh_per_hour": 60
  }
}
```

একই body এখানেও আছে <https://github.com/meherabmehu/GridWise-LLM-Energy-Optimizer/blob/main/docs/test-cases/request-bodies/m1.json>